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Practical 11

To use JavaScript **timers** (setTimeout() and setInterval()), implement **dynamic UI updates**, and organize code using **ES6 modules (import/export)** to create a countdown or class scheduler that highlights active sessions and adapts to viewport changes.

Requirements

Software / Tools

- Web Browser
- Code Editor
- Local server setup (VS Code Live Server or Node.js) — required for ES6 modules

Files

- index.html
- main.js (main script file)
- timer.js (module file)
- ui.js (module file)

Theory

1. Timers in JavaScript

JavaScript provides **built-in timer functions** that help execute code **after a delay** or **repeatedly at intervals**.

Method	Description	Example

setTimeout(fn, delay)
 	
setInterval(fn,
	Executes fn **once** after the given delay (in ms).
	
	Executes fn **repeatedly** at every given delay (in ms).
		setTimeout(() => console.log("Hello"), 2000);
		
		setInterval(() => console.log("Tick"), 1000);

delay)		
clearTimeout() / clearInterval()	Stops the timer created by the above methods.	clearInterval(timerId);

2. Dynamic UI

Dynamic UI refers to **real-time updates** of webpage content using JavaScript — such as:

- Updating time every second.
- Highlighting the active schedule session.
- Adjusting styles or layouts when the window is resized.

3. Event Delegation

Event delegation means attaching a **single event listener** to a **parent element** instead of each child. It uses the concept of **event bubbling**.

Example:

```
document.getElementById('list').addEventListener('click', (e) => {
  if (e.target.tagName === 'LI') {
    alert("You clicked " + e.target.textContent);
  }
});
```

4. ES6 Modules

Modules allow you to **split code into separate files** and **reuse** it easily.

Keyword	**Purpose**
export	Makes functions/variables available for use in other files.
import	Brings in code from another module.

Example:

```
// math.js
export function add(a, b) { return a + b; }

// main.js
import { add } from './math.js';
console.log(add(2, 3));
```

Note: You must run ES6 modules using a **server environment** — not just by opening HTML directly.

Steps

1. Create a folder Practical11_Timers_Modules.
2. Inside it, create these files:
    - index.html
    - main.js
    - timer.js
    - ui.js
3. Add a basic HTML structure and load the main script using type="module".
4. Implement timer and UI functions in separate modules.
5. Import them in main.js and call the functions.
6. Test in browser using VS Code Live Server or any local server.

Example Implementation

index.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <title>Timers, Dynamic UI & Modular Design</title>

  <!-- Basic styling for the page -->
  <style>
    body { font-family: Arial; padding: 20px; }
```

```
    /* Each class session block */
    .session {
      padding: 10px;
      margin: 5px 0;
      border: 1px solid #ccc;
      border-radius: 5px;
    }

    /* Highlight active session */
    .active {
      background: #c8f7c5;
    }
  </style>
</head>
<body>
  <h2>Class Scheduler (with Timers & Modules)</h2>

  <!-- Container for all sessions -->
  <div id="sessions">
    <div class="session" data-time="09:00">HTML Basics - 9:00 AM</div>
  <div class="session" data-time="10:00">CSS Design - 10:00 AM</div> <div
  class="session" data-time="11:00">JavaScript Logic - 11:00 AM</div>
  </div>

  <!-- Display countdown timer -->
  <p id="countdown"></p>

  <!-- Main entry JavaScript file (module-based) -->
  <script type="module" src="main.js"></script>
</body>
</html>
```

timer.js

```
// timer.js — contains countdown and session highlight logic

// Function to start a countdown timer
export function startCountdown(duration, display) {
  let time = duration; // time in seconds

  // setInterval() runs the code every 1 second
  const timer = setInterval(() => {
    const minutes = Math.floor(time / 60); // convert seconds to minutes
    const seconds = time % 60; // remaining seconds

    // Update countdown message on the page
    display.textContent = `Next session starts in ${minutes}:${seconds < 10 ? '0' + seconds
: seconds}`;

    time--; // decrease the timer each second

    // Stop timer when it reaches 0
    if (time < 0) {
      clearInterval(timer);
      display.textContent = "Session Started!";
    }
  }, 1000); // executes every 1000 milliseconds (1 second)
}


// Function to highlight the currently active session
export function highlightActiveSession() {
  const sessions = document.querySelectorAll('.session'); // get all session elements
  const now = new Date(); // get current date/time
  const currentHour = now.getHours(); // extract current hour (24-hour format)

  // Loop through all session divs
  sessions.forEach(session => {
    const hour = parseInt(session.dataset.time.split(':')[0]); // extract hour from data-time
```

```
    // If current hour matches session time → highlight
    if (hour === currentHour) {
      session.classList.add('active');
    } else {
      session.classList.remove('active');
    }
  });
}
```

ui.js

```
// ui.js — handles dynamic UI and responsive design behavior
// Function to change background color based on screen size
export function handleViewportChange() {
  // If screen width < 600px, apply a light blue background
  if (window.innerWidth < 600) {
    document.body.style.background = '#f0f8ff';
  } else {
    // For larger screens, use white background
    document.body.style.background = 'white';
  }
}


// Function to add dynamic behavior (event delegation)
export function addDynamicSessions() {
  const sessionsDiv = document.getElementById('sessions');

  // Add event listener to parent container (event delegation)
  sessionsDiv.addEventListener('click', (e) => {
    // Check if clicked element has class "session"
    if (e.target.classList.contains('session')) {
      alert("You selected: " + e.target.textContent);
    }
```

```
  });
}
```

main.js

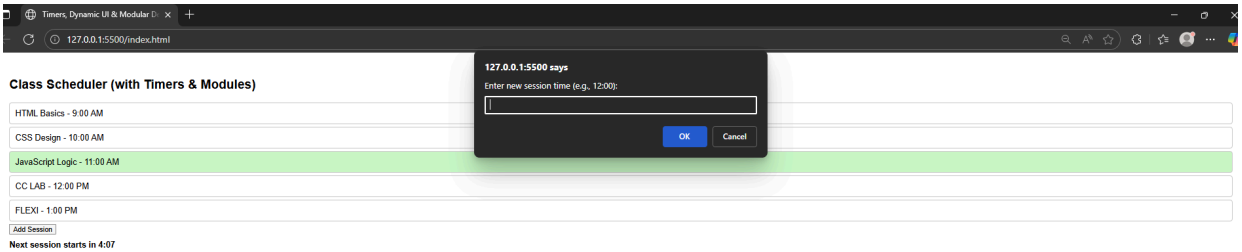
```
// Import functions from timer.js and ui.js modules
import { startCountdown, highlightActiveSession } from './timer.js';
import { handleViewportChange, addDynamicSessions } from './ui.js';

// Select the countdown display element from the DOM const
countdownDisplay = document.getElementById('countdown');

// Start countdown for next class (e.g., 5 minutes = 300 seconds)
startCountdown(300, countdownDisplay);

// Highlight the active session every minute automatically
setInterval(highlightActiveSession, 60000); // recheck every 60 seconds
highlightActiveSession(); // run once immediately on load

// Handle screen resizing for responsive behavior
window.addEventListener('resize', handleViewportChange);
handleViewportChange(); // run on page load as well

// Enable event delegation on session elements
addDynamicSessions();
```

Output

- Displays **class schedule**.
- Shows a **countdown timer** (updates every second).
- Automatically **highlights the current class session** based on system time.
- Background color **changes** when window is resized (responsive UI). ● Clicking a session triggers an **alert** (event delegation).

Task

1. Add a **new session dynamically** using a button (“Add Session”).
2. Modify the countdown so that it changes color when time < 1 minute.
3. Use a separate module helper.js to format time or log events.
4. Add responsiveness — change font size when width < 500px.



TIME REMAINING < 1 min:-